

Network Hardware

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Reflecting ideas by:

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High-level message

- Hardware innovations play a critical role in all types of networks
- The networking community has made significant strides in software stacks and applications
- But historically our footprint in hardware design has been smaller
- This is not due to lack of interest from the research community
 - Think Tofino

Research themes in network hardware

1. Programmable hardware
2. Integration of hardware with applications
3. Network infrastructure/testbeds/prototypes
4. The rise of machine learning and its impact on networking hardware
5. Fostering innovation in emerging hardware for the networking community

Programmable Hardware

- What is the future roadmap for programmable network hardware?
 - Explore new designs for programmable packet processing pipelines
 - Think beyond RMT and Tofino, e.g., Juniper Trio
 - Make stateful processing a first-order objective
 - Identify new and emerging applications that can leverage programmable network hardware
 - IoT analytics / Edge computing, AR/VR
 - Explore new programmable primitives
 - Application-guided primitive design
 - Push for programmable hardware virtualization
 - Dynamic resource allocation
 - Time-share hardware resources between multiple applications
 - Add/remove applications without switch downtime
 - Extend programmability beyond switches and towards the edge network devices
 - SmartNICs with programmable processing pipelines

Integration of hardware with applications

- How to design and test new hardware devices together with applications
 - Meaningful performance metrics
 - Workloads to evaluate
 - FPGA vs. P4 vs. emulation vs. simulation vs. asic synthesis
 - How to evaluate at scale
 - Programmability, compilers, abstractions

The rise of machine learning

- The network is a bottleneck for DNN training and inference
- We need to think about ML as a special workload
 - Software and hardware co-design, specialized co-processors
 - Need transport libraries, congestion control, virtualization
 - Training, inference, large-language models, mixture of experts,
 - In-network acceleration
- Power/energy perspective
 - Training ChatGPT requires millions of GPU hours
 - Serving ChatGPT inference for a month consumes millions of KWh
 - PCIe physical layer has a much larger energy footprint

Networking Infrastructure

- Networking infrastructure/testbeds is vital
 - CloudLab, FABRIC, Chameleon, San Diego Supercomputer Center (SDSC), ...
 - Unsynchronized SW/HW ecosystems
 - System integration difficulty
- High-barrier to entry
 - How to share hardware/prototypes/testbed?
 - Not only the hardware platform, but also the software stacks, lessons, experiences
- How to facilitate starting a new hardware-oriented research agenda?

Fostering innovation in emerging hardware

- Co-design networking solutions with emerging hardware
 - Optics:
 - Communication: novel fiber technologies
 - Multi-chip modules: Co-packaged optics/Silicon photonics:
 - Computation: analog photonic computing
 - CXL
 - Rethink the node and rack architecture
 - Not just memory, but also devices
 - Closer integration and cache-coherent interconnects of accelerators
 - Node participating in multiple networks
 - Novel switch design
 - NIC design

Summary

- Research themes in network hardware:
 - Programmable hardware
 - Integration of hardware with applications
 - Network infrastructure/testbeds/prototypes
 - The role of Machine learning and AI in networking hardware
 - Fostering innovation in emerging hardware for the networking community
- Additional questions:
 - Which applications benefit the most from hardware innovations? ML, gaming, metaverse, ...
 - What kind of testbed/hardware the research community needs?
 - Quantum networking? Is that even a hardware problem? Is it relevant in the current timeline?